

Seminar Arbeit

From the Gai Kapadia model to its empirical verification

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*Are more connections increasing the stability of the system ?
From the Gaussian point of view to the Power Law*

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1 Introduction

The modern financial system is defined by a dense web of interconnected institutions whose mutual dependencies span the entire global economy. While this complexity was long considered a source of resilience — allowing risk to be distributed across many participants — the crisis of 2008 exposed its darker counterpart: that the same channels designed to absorb shocks can just as readily transmit them. The collapse of Lehman Brothers did not remain a local event; it propagated across balance sheets, asset classes, and national borders, demonstrating that systemic risk is fundamentally a collective phenomenon rather than the sum of individual failures. Understanding the mechanisms by which distress spreads through financial networks has since become a central preoccupation of both academic research and prudential regulation.

This paper builds upon the structural analysis of financial networks introduced by Clément Landry [1], and asks a sharper question: is the Gai-Kapadia model empirically correct? To answer this, we examine in depth the model proposed by Gai and Kapadia [2], which offers a rigorous and analytically tractable account of contagion in interbank networks, and confront its assumptions with evidence from real interbank markets.

The central question we set out to address is the following: does increased connectivity genuinely enhance the stability of a financial system? The Gai-Kapadia framework, we argue, rests implicitly on the assumption that interbank exposures follow a Gaussian distribution, arising naturally from the use of Erdős-Rényi random graphs. This assumption renders losses self-averaging and crises statistically improbable — yet, when they occur, catastrophic. Empirical evidence, however, suggests that real interbank networks are better described by Power Law distributions and scale-free topologies, a distinction with profound consequences for how risk concentrates and how it propagates.

The paper proceeds as follows. Section 2 introduces the necessary tools from graph theory and establishes the network-theoretic language used throughout. Section 3 examines the statistical contrast between Gaussian and Power Law shock distributions, and motivates why this distinction is not merely technical but determinative of the system’s qualitative behavior. Section 4 presents the Gai-Kapadia model in detail, with particular attention to its non-linear dynamics, phase transition structure, and the convergence properties of its finite-network simulations. Section 5 compares the model’s predictions against the empirical characteristics of real financial networks. Finally, Section 6 draws together the policy implications of our analysis and discusses the lessons that the 2008 crisis, and the subsequent Basel III reforms, offer for the calibration of theoretical models.

2 Network and Graph Theory

2.1 From Isolation to Interdependence

As established in the introduction, the 2008 financial crisis demonstrated that systemic risk cannot be understood by examining institutions in isolation. Banks borrow, lend, and share risk through dense webs of bilateral contracts, and it is precisely these linkages that transform a local failure into a global crisis. To model this structure rigorously, we turn to the language of graph theory, which provides both the vocabulary and the analytical tools needed to study how distress propagates through a financial system.

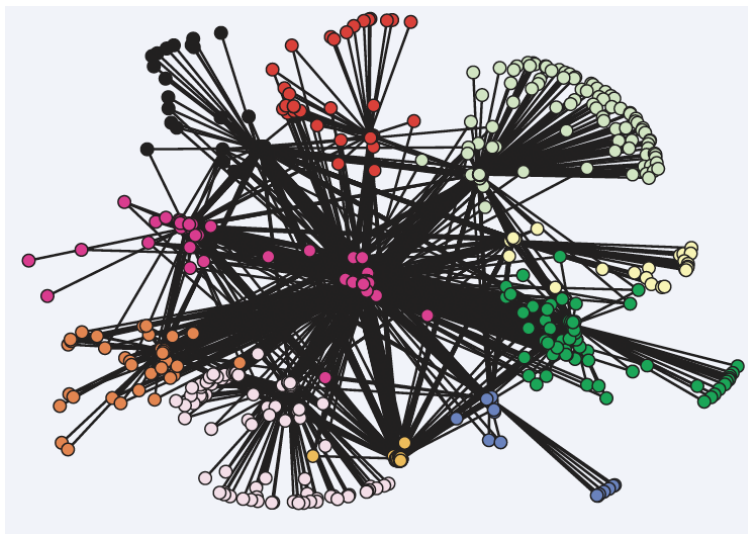


Figure 1: The Austrian interbank network [3]. Each node represents a financial institution, and each directed edge represents an exposure. The heterogeneous degree distribution — with a few highly connected hubs and many peripheral nodes — is characteristic of real-world financial networks and motivates the Power Law analysis developed in Section 3.

2.2 Graph Theory Basics

We represent a financial system as a directed graph $G(V, E)$, where:

- **Vertices (Nodes) V :** represent economic agents such as banks, insurance companies, or households.
- **Edges (Links) E :** represent financial relationships between agents, such as loans, ownership stakes, or common asset holdings.

The connectivity structure of the graph is encoded in the **adjacency matrix** A , defined by:

$$A_{ij} = \begin{cases} 1 & \text{if there is a directed link from node } i \text{ to node } j \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

As a concrete example, consider a network of five nodes with the following adjacency matrix and edge set:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}, \quad V = \{1, 2, 3, 4, 5\}, \quad E = \{1 \rightarrow 2, 3 \rightarrow 4, 5 \rightarrow 2, 4 \rightarrow 5, 3 \rightarrow 5\}$$

While the in-degree of node i , defined as $k_i^{\text{in}} = \sum_j A_{ji}$, counts the number of creditors that have an exposure to i , the out-degree $k_i^{\text{out}} = \sum_j A_{ij}$ counts the number of counterparties to which node i is exposed.

2.3 Types of Financial Relations

Financial links vary considerably in nature, and recognizing this heterogeneity is essential for understanding both the richness of real interbank systems and the simplifications that theoretical models must make. We distinguish four levels of complexity.

Directed and undirected graphs. A loan from commercial bank i to commercial bank j creates an asymmetric exposure: i bears the credit risk of j , but not vice versa. This is naturally represented as a **directed** graph, where $A_{ij} \neq A_{ji}$ in general. By contrast, a symmetric relationship such as a joint investment is better modeled as an **undirected** graph, in which $A_{ij} = A_{ji}$.

Weighted graphs. In most financial applications, the mere existence of a link is insufficient; its monetary value matters. We therefore assign a weight $\omega_{ij} \in \mathbb{R}_+$ to each edge, representing the size of the exposure. The binary adjacency matrix A is then replaced by a weighted matrix W with entries $W_{ij} = \omega_{ij}$, which allows the model to distinguish a small interbank loan from a systemically significant one.

Different types of actors. Real financial networks are not populated by identical institutions. A central bank occupies a structurally different position from a commercial bank: it acts as a lender of last resort, holds unique regulatory powers, and its failure — unlike that of a commercial bank — is not modeled as a credit event. Similarly, insurance companies, hedge funds, and clearing houses each interact with the network in distinct ways. Recognizing heterogeneity in node types is therefore important for accurately representing who can fail and how.

Different types of relations. Two institutions may be connected simultaneously through multiple channels: a direct lending relationship, a shared equity stake, and an exposure

through derivatives contracts. Each type of relation transmits risk differently and responds differently to regulatory intervention. A complete model would represent each relation type as a separate layer of the network.

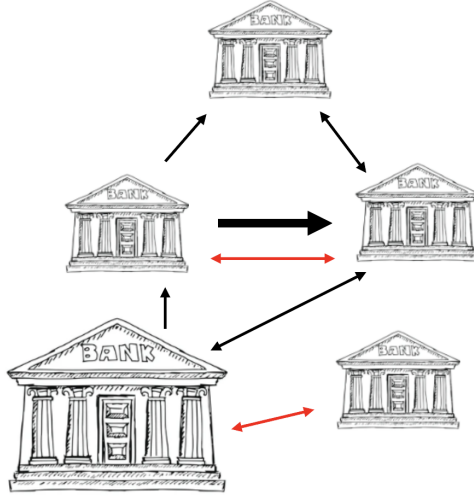


Figure 2: Illustration of a heterogeneous financial network. Arrows indicate directed credit exposures: a directed edge from i to j means that institution i holds a claim on j and bears the risk of j 's default. Different edge colors represent distinct types of contractual relationship. Node size distinguishes institution types: larger nodes represent central banks or clearing houses, smaller nodes represent commercial banks.

Note: The Gai-Kapadia model retains only the first two levels of this complexity. It represents the interbank system as a **weighted directed graph** with a single type of relationship (interbank lending) and a single type of actor (commercial banks). Central banks, derivatives, and equity cross-holdings are all abstracted away. This simplification makes the model analytically tractable, but it also means that the heterogeneity documented empirically in Section 5 lies entirely outside its scope.

2.4 Quantitative Tools for Contagion

To characterize how vulnerable a network is to shock propagation, we rely on several structural metrics.

Degree distribution P_k : the probability that a randomly selected node has exactly k connections. The shape of this distribution is critical: a highly concentrated distribution, with a few nodes of very high degree, indicates a system in which the failure of a single hub can immediately affect a large fraction of the network. As we will see in Section 5, this is precisely the topology observed in real interbank markets.

Average degree z : in a directed graph, the mean in-degree equals the mean out-degree, and both are equal to the average number of counterparties per institution:

$$z = \sum_{j,k} j \cdot P_{jk} = \sum_{j,k} k \cdot P_{jk} \quad (2)$$

This parameter plays a central role in the Gai-Kapadia model, where it serves as the primary control variable for the onset of systemic contagion.

Clustering coefficient C : the probability that two neighbors of a given node are also neighbors of each other. A high clustering coefficient implies that failures tend to be concentrated within densely connected subgraphs, which can amplify local shocks before they spread to the rest of the system.

Average path length and the small-world property: the average path length $\langle \ell \rangle$ measures the mean number of steps required to travel between any two nodes. Networks in which $\langle \ell \rangle$ is small relative to the total number of nodes are called *small-world* networks. In such systems, a distress signal can reach any institution in only a few steps, meaning that local shocks become global almost instantaneously.

2.5 A Physics Perspective

The network-theoretic framework described above has a natural interpretation in terms of statistical mechanics, which we find particularly illuminating. The propagation of financial distress through an interbank network is formally analogous to a **percolation problem** [4]: nodes fail sequentially as the stress they face exceeds their capital buffer, and the question is whether the cascade reaches a macroscopic fraction of the system.

More precisely, the transition from a stable regime — in which defaults remain localized — to a systemic crisis is a **phase transition** in the sense of statistical mechanics. A single control parameter, the average degree z , plays the role of temperature: as it crosses a critical threshold, the global state of the system changes discontinuously. We will formalize this analogy in Section 4, where the phase diagram of the Gai-Kapadia model is derived explicitly.

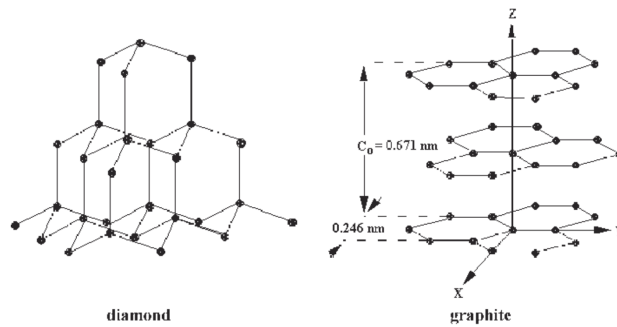


Figure 3: Crystal structures of diamond (left) and graphite (right) as an analogy for financial network topology. In diamond, each carbon atom is connected to four others in a rigid three-dimensional lattice, creating a highly resilient structure. In graphite, atoms are strongly connected within layers but weakly connected across them, creating a direction-dependent fragility [5]. Similarly, the topology of interbank connections — not merely their number — determines whether a financial network absorbs or amplifies a shock.

3 Gaussian vs. Power-Law: The Nature of Shocks

The architecture of a financial network determines the pathways along which distress can travel, but the outcome of a crisis depends equally on the nature of the initial shock. Before analyzing the mechanics of contagion in the Gai-Kapadia model, we must therefore ask a more fundamental question: what is the statistical character of the shocks to which a financial system is exposed? The answer to this question — whether extreme events are exponentially rare or structurally expected — has profound consequences for how we assess and regulate systemic risk.

3.1 Two Worlds: Mediocristan vs. Extremistan

Following the terminology introduced by Taleb [6], we distinguish between two statistical regimes that lead to qualitatively different pictures of financial stability.

In the **Gaussian world** (thin tails), observations are concentrated around the mean and large deviations are exponentially suppressed. The system has a well-defined typical scale, and extreme events are so unlikely as to be practically impossible: a 10σ shock, for instance, would not be expected to occur even once over the lifetime of the universe. Risk in this regime is, in principle, manageable.

In the **Power-Law world** (heavy tails), the distribution decays slowly as a polynomial. Large shocks are rare, but they are not negligible — they are structurally built into the distribution. Crucially, if the tail exponent α satisfies $\alpha \leq 2$, the variance of the distribution is infinite, meaning that the system has no characteristic scale for disasters. Extreme events cannot be treated as anomalies; they must be treated as an expected feature of the environment.

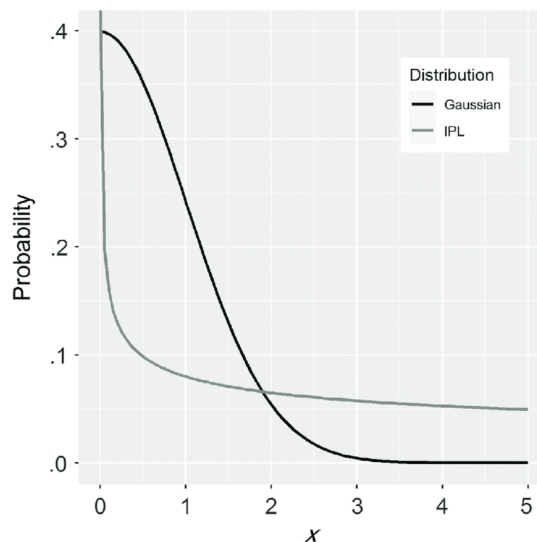


Figure 4: Probability density functions of a Gaussian distribution and an inverse power law (IPL). The Gaussian curve falls off exponentially beyond the mean, while the power-law tail decays slowly, assigning non-negligible probability to very large events. This difference in tail behavior is the primary driver of the contrasting stability properties discussed throughout this paper.

3.2 Mathematical Comparison

The two regimes are distinguished by the form of their probability density functions (PDFs). Table 1 summarizes the key properties.

Feature	Gaussian (Normal)	Power Law (Pareto)
Equation	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$f(x) = Cx^{-\alpha}$
Mean $\mathbb{E}[X]$	μ	finite ($\alpha > 1$), ∞ ($\alpha \leq 1$)
Variance $\text{Var}(X)$	σ^2	finite ($\alpha > 2$), ∞ ($\alpha \leq 2$)
Tail decay	Exponential (rapid)	Polynomial (slow)
Extreme events	Negligible	Structurally present

Table 1: Comparison of the Gaussian and Power Law (Pareto) distributions. Threshold values for the mean and variance refer to the PDF parameterization $f(x) = Cx^{-\alpha}$; the survival-function convention $P(X \geq x) \propto x^{-\alpha}$ shifts these thresholds by one [7].

The choice of distributional assumption is not a technical detail; it determines the qualitative behavior of the entire system. A model calibrated under Gaussian assumptions will systematically underestimate the probability of tail events, and therefore the true extent of systemic risk. As we will see, this is precisely the limitation of the Gai-Kapadia framework.

3.3 Implicit Gaussianity in the Gai-Kapadia Model

The Gai-Kapadia model represents the interbank network as an **Erdős-Rényi random graph**, in which each pair of nodes is connected independently with probability p . For a network of N banks, the degree of each node then follows a Poisson distribution with mean $\lambda = (N - 1)p$. By the Central Limit Theorem, when λ is large, this Poisson distribution converges to a Gaussian, so that the model effectively operates in a thin-tailed regime.

Three consequences follow directly from this construction. First, no institution accumulates a disproportionate number of counterparties: the degree distribution has an exponentially decaying tail, so that nodes with anomalously many connections are vanishingly unlikely. Second, interbank exposures are assumed to be independent and identically distributed (i.i.d.), so that the loss incurred by a bank upon the default of a neighbor is drawn from the same distribution regardless of which neighbor fails. Third, and most importantly, the aggregate loss faced by any given institution is a sum of i.i.d. random variables with finite variance. The CLT then guarantees that this sum is asymptotically Gaussian, making total losses *self-averaging*: in the large- N limit, the realized loss concentrates tightly around its expectation (see Appendix A.2).

This self-averaging property is what makes the mathematics of the GK model tractable and

elegant. It also, as we will argue in Section 5, makes it structurally blind to the fat-tailed reality of modern interbank markets, where the failure of a single highly connected institution can dominate the loss distribution of the entire system (see Appendix A.3 for the contrasting Power-Law case).

4 The Gai-Kapadia Model: Robust yet Fragile

The seminal paper by Gai and Kapadia [2] provides a cornerstone for the analytical study of financial contagion. Their central result is a paradox that challenges the naive intuition that more connectivity is always better: financial networks can be simultaneously **robust yet fragile**. The probability that any given shock triggers a systemic crisis may be low, but when contagion does occur, it is catastrophic in extent. Understanding the precise conditions under which this transition happens is the main objective of this section.

4.1 Modeling the Agent: The Bank Balance Sheet

The fundamental unit of the model is a bank, represented by its balance sheet. On the asset side, a bank holds outside assets (illiquid investments) and interbank assets (claims on other institutions). On the liability side, it carries outside liabilities (deposits and wholesale funding) and interbank liabilities (debts owed to counterparties), with the residual defined as net worth w_i .

A bank i defaults when its net worth falls to zero, that is, when the losses it sustains from the default of its counterparties exceed its capital buffer c_i . Formally, if a fraction q of its z neighbors default and each bilateral exposure has value $\bar{p}$, bank i defaults if:

$$q \cdot z \cdot \bar{p} > c_i \quad (3)$$

This simple condition encodes the entire contagion mechanism: a bank survives if its capital is sufficient to absorb the losses from its defaulted counterparties, and fails otherwise, potentially triggering further defaults downstream.

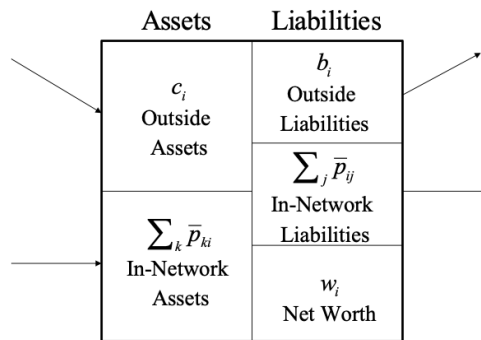


Figure 5: Stylized balance sheet of a bank in the Gai-Kapadia model. Outside assets and interbank claims appear on the left; outside liabilities, interbank obligations, and net worth on the right. Default occurs when accumulated losses from counterparty failures drive net worth w_i to zero.

4.2 Connectivity and the Topology of Risk

The model is governed by three key quantities: the average degree z (mean number of counterparties per bank), the *frequency* Y (the probability that a single initial failure triggers a systemic crisis, defined as more than 5% of all banks failing), and the *extent* (the total fraction of banks that ultimately default, conditional on a crisis occurring). Together, frequency and extent capture both how likely a crisis is and how severe it will be.

Numerical simulations of the model reveal that contagion is confined to a specific window of connectivity, which we can decompose into four qualitatively distinct regimes.

At low connectivity ($z < 2$), the network is too sparse for shocks to propagate. A defaulting bank has so few counterparties that its failure does not generate sufficient losses elsewhere to trigger secondary defaults, and the distress remains localized.

As connectivity increases toward $z \approx 3$, the network enters what we call the *danger zone*. The system is now connected enough for a shock to travel across multiple hops, but not sufficiently diversified for individual banks to absorb the incoming losses. This is the regime of maximal contagion frequency: a single failure is most likely to cascade into a systemic crisis here.

At strong connectivity ($z > 5$), the picture reverses. Each bank now shares its interbank exposures across many counterparties, so the loss attributable to any single default is small relative to the receiving bank's capital buffer. Risk-sharing dominates, and the system becomes robust to small shocks.

Finally, beyond a critical threshold (indicated by the dashed line in Figure 6), contagion disappears entirely from the simulations. It is important to interpret this correctly: the drop of *extent* to zero does not mean that crises become smaller. It means they cease to occur altogether, leaving the extent undefined. The system has crossed from the supercritical into the subcritical phase.

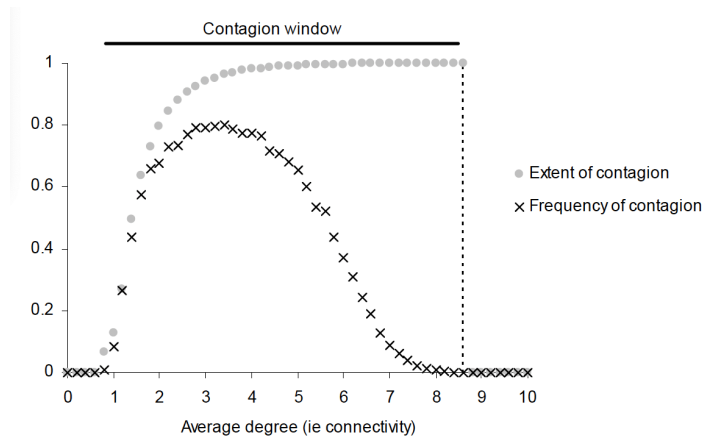


Figure 6: Frequency and extent of contagion as a function of average degree z , from Gai and Kapadia [2]. The frequency of contagion peaks near $z \approx 3$ before declining, while the extent remains high within the contagion window. Beyond the dashed threshold, both quantities drop to zero as the system enters the subcritical regime.

4.3 Non-Linearity, Phase Transitions, and the Knife-Edge Property

Two features of the model deserve particular emphasis, as they have direct consequences for financial regulation.

The first is strict non-linearity. The relationship between connectivity and stability is not monotone: increasing z from 2 to 3 does not improve network safety but in fact maximizes the probability of global collapse. Standard regulatory intuitions — that more diversification is always better, or that incremental changes in lending requirements have proportional effects — are fundamentally misleading in this setting.

The second is what Gai and Kapadia call the *knife-edge* property. Two initial shocks that are statistically indistinguishable can lead to radically different outcomes. At $z \approx 8$, for instance, an identical shock may have zero systemic effect in one simulation run, or trigger the total collapse of every institution in the network in another. The system’s response is exquisitely sensitive to microscopic details of the network realization, making it essentially impossible to predict or prevent individual crises using conventional oversight tools.

From the perspective of statistical mechanics, this behavior is immediately recognizable as a **phase transition**. The average degree z plays the role of a control parameter — analogous to temperature in a thermodynamic system — and the onset of systemic contagion corresponds to the system crossing a critical point. Just as water does not gradually become ice but undergoes a discontinuous change of state at a sharp temperature threshold, the financial network transitions abruptly from a stable subcritical phase to a contagion-dominated supercritical phase as z crosses its critical value. The transition is not smooth; it is structural.

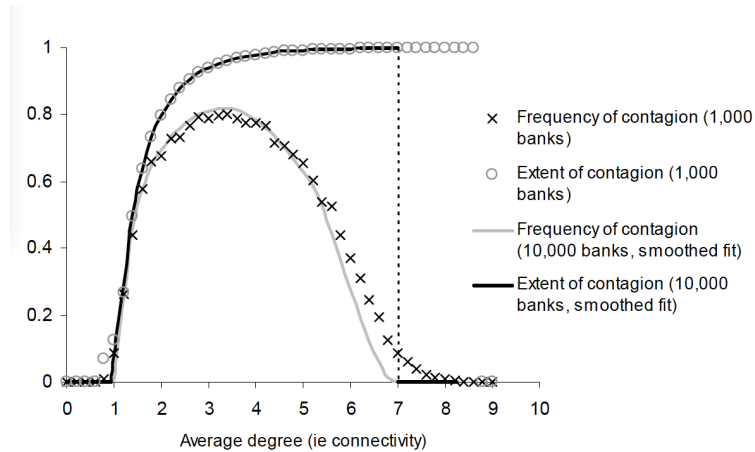


Figure 7: Comparison of contagion simulations for $N = 1,000$ and $N = 10,000$ banks. The near-identical results confirm that the robust-yet-fragile property is a structural feature of the network topology, not a finite-size artifact. As $N \rightarrow \infty$, the Gaussian self-averaging argument of Section 3.3 guarantees convergence to a deterministic limit.

4.4 Structural Convergence and the $N \rightarrow \infty$ Limit

A natural concern about any simulation-based result is whether it reflects a genuine property of the model or merely an artifact of the finite system size. The Gai-Kapadia results are robust in this respect. As established in Section 3.3, the loss distribution in the GK model is asymptotically Gaussian, and the Central Limit Theorem guarantees that aggregate losses concentrate around their mean as $N \rightarrow \infty$. The robust-yet-fragile property is therefore not a numerical accident but a structural consequence of the network’s geometry.

This convergence is confirmed directly by comparing simulations at $N = 1,000$ and $N = 10,000$ banks, as shown in Figure 7. The frequency and extent curves are nearly indistinguishable across the two system sizes, establishing that the contagion window and the phase transition threshold are intrinsic properties of the interbank topology rather than finite-sample noise. The risk, in other words, is embedded in the geometry of the links themselves.

5 Empirical Validation: From Mediocristan to Extremistan

The Gai-Kapadia model offers an analytically tractable account of financial contagion, but its predictive value ultimately depends on how faithfully it represents the structure of real interbank markets. In this section, we confront the model’s Gaussian assumptions with empirical evidence from observed financial networks. The comparison reveals a fundamental mismatch: while the GK framework assumes a homogeneous, Erdős-Rényi topology, real interbank markets exhibit the heterogeneous, hub-dominated structure characteristic of scale-free networks. This distinction is not merely topological — it changes the qualitative nature of systemic risk.

5.1 The Gaussian Perspective: Risk as a Statistical Average

In the Gai-Kapadia framework, the financial system operates under the Law of Large Numbers. Because exposures are i.i.d. and the degree distribution is Poisson, aggregate risk is well-described by its mean, and the behavior of any individual institution is representative of the whole. Stability in this world arises from diversification: by spreading exposures across many similar counterparties, each bank reduces the impact of any single default to a manageable fraction of its capital buffer.

Fragility, correspondingly, is a threshold phenomenon. The system remains stable as long as the aggregate volume of shocks stays below the critical level required to exhaust the network’s collective capital. Once that threshold is crossed, the phase transition described in Section 4.3 occurs, and defaults cascade globally. Below the threshold, however, the system is genuinely robust: shocks are absorbed by being spread thin across the network, and no single institution’s failure is large enough to destabilize the whole.

This picture is internally consistent and mathematically elegant. Its limitation is that it assumes away the very feature of real financial networks that makes them most dangerous: extreme heterogeneity in the distribution of connections and exposures.

5.2 The Power-Law Perspective: Risk as a Structural Property

Empirical studies of interbank networks consistently find that the degree distribution and the distribution of bilateral exposure sizes follow a Power Law rather than a Poisson distribution [8, 3]. Figures 8, 9, and 10 display log-log plots of contract size distributions and degree distributions from real interbank data. The approximately linear relationship on the log-log scale is the signature of a power law: $P(k) \sim k^{-\alpha}$, with estimated exponents in the range $\alpha \approx 1$ to 3 depending on the market and the quantity measured.

In a scale-free network, the notion of an “average” institution loses its meaning. The system is instead defined by its hubs: a small number of highly connected institutions that intermediate a disproportionate share of total interbank activity. Major institutions of this type occupy a structurally different position from the periphery, and their behavior cannot be inferred from the mean.

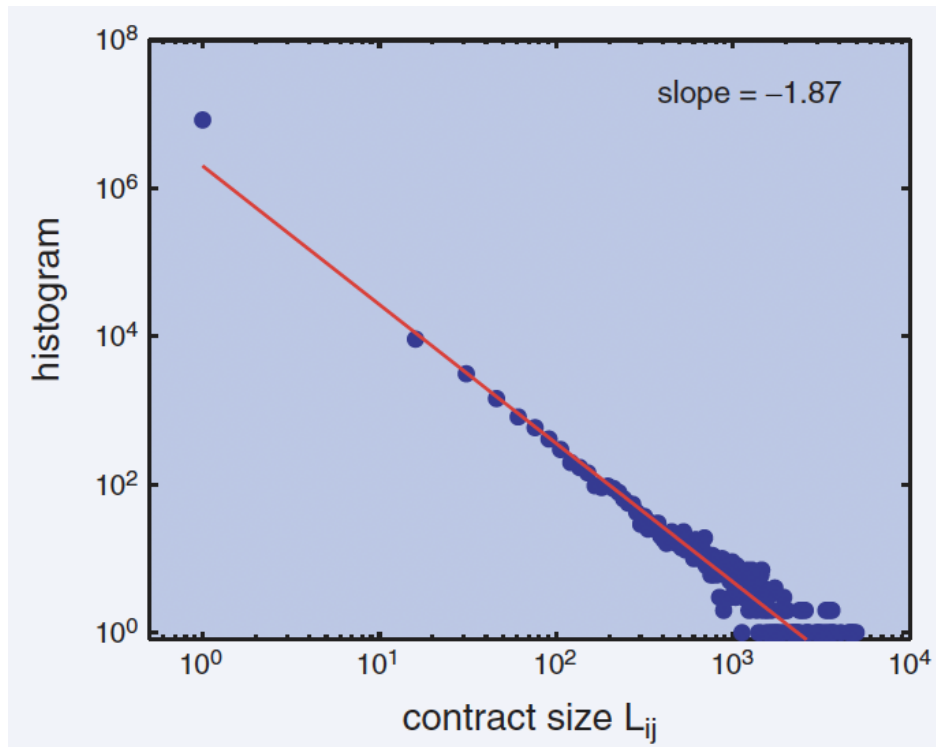


Figure 8: Log-log histogram of bilateral contract sizes L_{ij} in an empirical interbank network. The approximately linear slope of -1.87 on the log-log scale confirms a power-law distribution of exposure sizes, implying that extremely large bilateral contracts — absent from Erdős-Rényi models — are a structural feature of real markets.

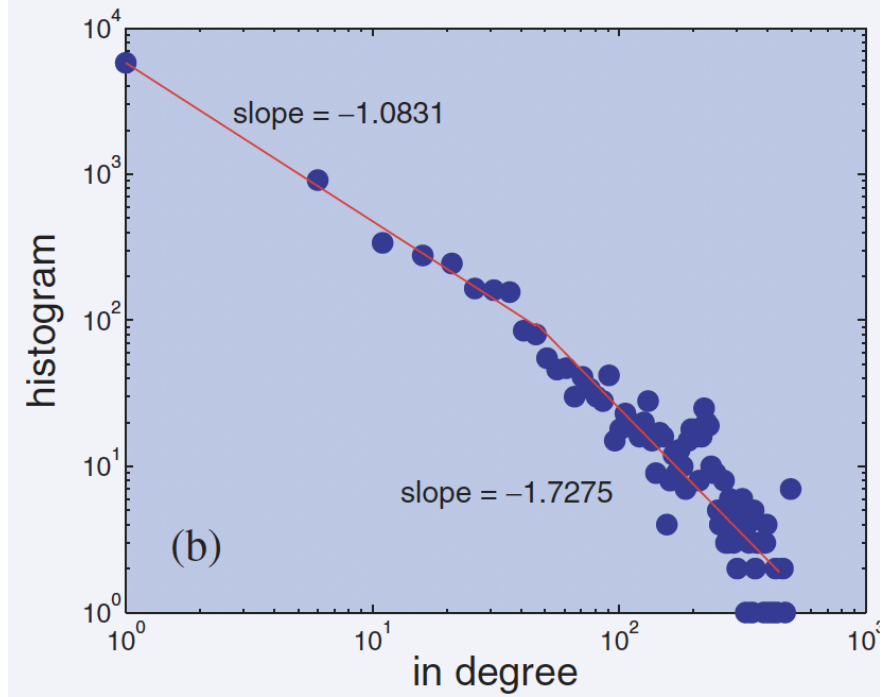


Figure 9: Log-log histogram of in-degrees (number of creditors) in the empirical interbank network. The two-regime power-law fit, with slopes -1.08 for small degrees and -1.73 for large degrees, indicates the presence of a small number of institutions with a disproportionately large number of creditors — the hubs of the lending network.

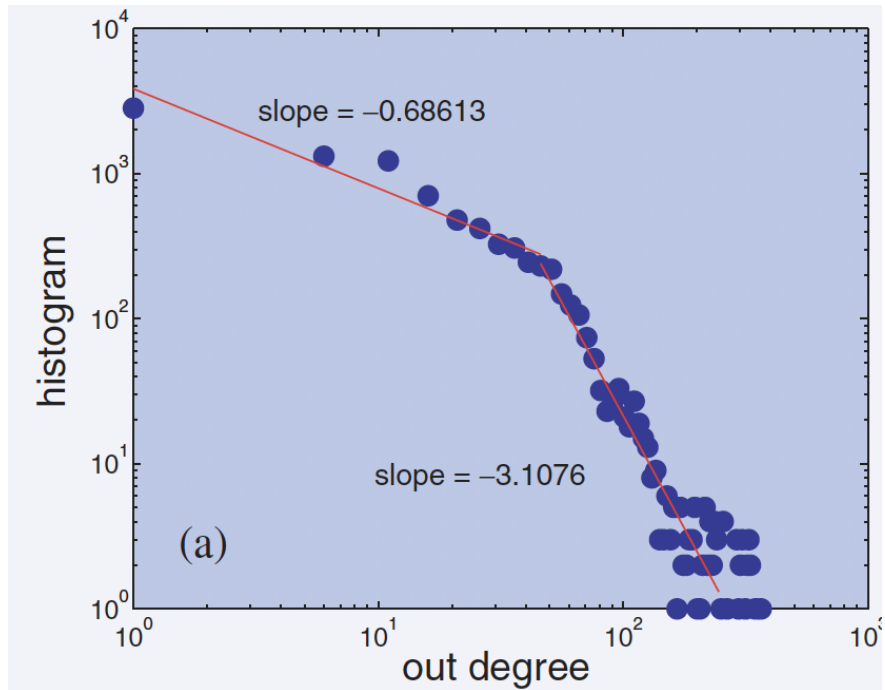


Figure 10: Log-log histogram of out-degrees (number of debtors) in the empirical interbank network. The slope of -0.69 in the bulk and -3.11 in the tail reflects a highly skewed distribution of borrowing activity, with a small number of institutions acting as dominant borrowers.

This topology generates a fundamentally different pattern of robustness and fragility. Against random failures — the default of a randomly selected bank — scale-free networks are remarkably resilient, because the vast majority of nodes are peripheral and their failure has negligible systemic impact. Against targeted failures, however, the network is acutely vulnerable. The collapse of a single hub propagates instantly to the large number of institutions directly exposed to it, and the resulting cascade can dominate the loss distribution of the entire system.

A further consequence is that the supercritical phase — in which a path to global contagion exists — is not confined to a window of intermediate connectivity, as in the GK model. Because hubs maintain permanent high-degree connections throughout the network, the topological conditions for systemic propagation are structurally present at all times. The financial system is, in this sense, always on the edge of criticality.

5.3 Summary Comparison

Table 2 summarizes the structural differences between the two frameworks.

Feature	Gaussian World (GK)	Power-Law World (Scale-Free)
Network topology	Erdős-Rényi (homogeneous)	Scale-free (hub-dominated)
Robustness mechanism	Diversification across many similar counterparties	Absorption capacity of high-degree hubs
Fragility mechanism	Aggregate shock volume exceeds capital threshold	Targeted failure of a single hub
Shock propagation	Distributed and attenuated by spreading	Amplified at hubs; absorbed at periphery
Supercritical phase	Confined to intermediate connectivity window	Structurally present at all connectivity levels
Physics analogy	Sharp phase transition (discontinuous)	Permanent criticality (self-organized)

Table 2: Structural differences in systemic risk between the Gaussian (Gai-Kapadia) and Power-Law (scale-free) frameworks.

5.4 The Central Lesson

The comparison above points to a conclusion that is both mathematically precise and practically important. In a Gaussian world, systemic risk is described by the mean: the behavior of the average institution is representative of the whole, and risk management reduces to controlling aggregate exposure volumes. In a Power-Law world, risk is dominated by the extremes: the fate of the system rests on the tail of the distribution, and the behavior of a handful of dominant institutions is more consequential than the combined activity of all the rest.

Systemic risk is therefore not simply a scaled-up version of idiosyncratic risk. It is a structural property of the network, determined by the topology of connections rather than the average characteristics of individual nodes. A regulatory framework that monitors only the “average” bank — or that assumes exposures are drawn from a thin-tailed distribution — is structurally blind to the most dangerous configurations that real financial networks can exhibit.¹

¹For an accessible visual treatment of why power laws are pervasive in complex systems, see Muller, D. [9]. While not a primary academic source, it provides useful geometric intuition for the hub-and-spoke structure discussed here.

6 Conclusion and Outlook

6.1 Final Synthesis: The Connectivity Paradox

For decades, the prevailing economic intuition was that increasing connectivity would necessarily make the financial system more stable through better risk-sharing. Our analysis of the Gai-Kapadia model and Power-Law dynamics challenges this simplistic view.

While connectivity improves stability **locally** by allowing banks to diversify their exposures, it creates **systemic fragility** once certain topological thresholds are crossed. This highlights a critical lesson: economic mechanisms rarely behave in a linear fashion. More connectivity does not always imply more stability; instead, it can create "super-highways" for contagion.

6.2 Key Policy Implications

The results of this study suggest that monetary and regulatory policies must shift their focus from individual health to **network health**:

- **Network Supervision:** It is not enough to monitor a bank's internal balance sheet. Regulators must supervise the *structure* of the network to prevent the formation of dangerous contagion channels.
- **Accounting for Thresholds:** Policy-making must account for non-linear effects. A small change in interbank interest rates or lending requirements can push the system across a phase transition boundary, moving it from a stable state to a supercritical one.

6.3 Outlook: The Reality of Capital Buffers

The 2008 financial crisis provides a direct empirical test of the model's parameters. The capital buffer used in theoretical simulations of the GK model is typically set at approximately 4% — a figure that, in retrospect, proved inadequate to contain the cascading failures observed in practice. The collapse of Lehman Brothers, whose leverage left it with minimal effective capital against its interbank exposures, triggered precisely the domino dynamics the model predicts.

The regulatory response was the Basel III framework, which substantially raised minimum capital requirements relative to its predecessor. Where Basel II had set total capital requirements at approximately 8% of risk-weighted assets, Basel III raised the effective floor to roughly 13%, incorporating a capital conservation buffer and systemic risk surcharges for globally significant institutions[3, 10].

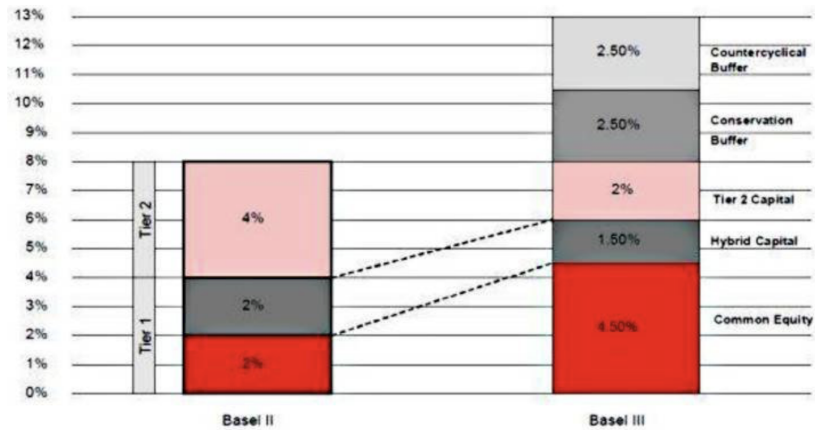


Figure 11: Capital requirement structure under Basel III [10]. The total requirement of approximately 13% is composed of a minimum Tier 1 capital ratio, a capital conservation buffer, and an additional surcharge for globally systemically important institutions (G-SIIs). Compared to the 8% floor under Basel II, this represents a structural shift in the regulatory approach to systemic risk.

In the language of the model, raising capital requirements corresponds to increasing the default threshold of each node: a larger shock is required to trigger any individual failure, which shifts the system away from the fragile intermediate connectivity regime identified in Section 4.2. To reduce systemic fragility, one must either reduce the density of dangerous connections through de-leveraging, or increase the resilience of the nodes themselves through higher capital. Basel III chose the latter path, and Figure 11 illustrates the resulting buffer structure.

The financial system is a complex dynamical system. Managing it effectively requires treating it as one — not as a collection of independent institutions, but as an interconnected network whose global properties emerge from the topology of its links. The tools of network theory and statistical mechanics, applied here through the lens of the Gai-Kapadia model, offer a rigorous foundation for that endeavor.

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A Mathematical Foundations of the Gaussian Regime

This appendix provides the mathematical foundations underlying the arguments developed in the main text. We first derive [11] the self-consistency equation that governs the onset of contagion in the Gai-Kapadia model, then prove that the loss distribution is asymptotically Gaussian in the Erdős-Rényi setting, and finally show why this Gaussian approximation breaks down when exposures follow a Power Law.

A.1 The Self-Consistency Equation and the Phase Transition

The fraction q of defaulted banks is not a free parameter of the model — it is determined endogenously by the network structure. A bank i defaults if the losses it sustains from its defaulted counterparties exceed its capital buffer c_i . Given that bank i has k_i counterparties and each bilateral exposure has normalized value $\bar{p}$, the default condition is:

$$\sum_{j=1}^{k_i} \mathbf{1}_{\{j \text{ defaults}\}} \cdot \bar{p} > c_i \quad (4)$$

Let q denote the probability that a randomly chosen bank has defaulted. On an Erdős-Rényi graph with average degree z , the number of defaulted neighbors of bank i follows a Binomial(k_i, q) distribution. For large k_i , this is approximated by Poisson(zq). Bank i defaults if the number of defaulted neighbors exceeds the threshold $\lfloor c_i/\bar{p} \rfloor$. Averaging over the degree distribution and the distribution of capital buffers, the fraction of defaulted banks satisfies the **self-consistency equation**:

$$q = \rho + (1 - \rho) \sum_{k=0}^{\infty} P_k \sum_{m > c/\bar{p}} \binom{k}{m} q^m (1 - q)^{k-m} \quad (5)$$

where ρ is the fraction of banks hit by the initial shock and P_k is the degree distribution. This equation may admit multiple solutions. A solution $q \approx \rho$ (close to the initial shock size) corresponds to the **subcritical phase**: contagion does not spread beyond the initially affected banks. A solution $q \gg \rho$ corresponds to the **supercritical phase**: a small initial shock propagates to a macroscopic fraction of the network. The onset of contagion — the phase transition identified in Section 4.3 — corresponds precisely to the bifurcation point at which a second, large solution to equation (5) first appears.

A.2 Formal Proof of the Gaussian Limit

We now prove that, under the Erdős-Rényi assumption, the individual loss L_i is asymptotically Gaussian and that aggregate losses are self-averaging.

Structural assumptions.

- The network is an Erdős-Rényi random graph. The degree k_i of each node follows a Poisson distribution with mean $\lambda = zp$:

$$P(k_i = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- Interbank exposures w_{ij} are i.i.d. with finite mean μ and finite variance σ^2 .
- Each neighbor defaults independently with probability q , so the per-neighbor loss variable is:

$$X_j = w_{ij} \cdot \mathbf{1}_{\{j \text{ defaults}\}}$$

Step 1: Moments of X_j . Since default occurs with probability q independently of the exposure size:

$$\mathbb{E}[X_j] = q\mu \tag{6}$$

$$\mathbb{E}[X_j^2] = q(\sigma^2 + \mu^2) \tag{7}$$

$$\text{Var}(X_j) = \mathbb{E}[X_j^2] - (\mathbb{E}[X_j])^2 = q\sigma^2 + q(1-q)\mu^2 \tag{8}$$

Step 2: Wald's Identity [12] for the compound sum. The total loss of bank i is a compound Poisson sum:

$$L_i = \sum_{j=1}^{k_i} X_j$$

where $k_i \sim \text{Pois}(\lambda)$ is independent of the X_j . Applying Wald's Identity:

$$\mathbb{E}[L_i] = \mathbb{E}[k_i] \mathbb{E}[X_j] = \lambda q\mu \tag{9}$$

For the variance, we use the law of total variance:

$$\text{Var}(L_i) = \mathbb{E}[k_i] \text{Var}(X_j) + (\mathbb{E}[X_j])^2 \text{Var}(k_i) \tag{10}$$

Substituting the Poisson property $\text{Var}(k_i) = \mathbb{E}[k_i] = \lambda$ and equation (8):

$$\text{Var}(L_i) = \lambda[q\sigma^2 + q(1-q)\mu^2 + (q\mu)^2] = \lambda[q\sigma^2 + q\mu^2] = \lambda q(\sigma^2 + \mu^2) \tag{11}$$

Step 3: CLT convergence for individual losses. Since L_i is a sum of i.i.d. terms X_j with a random number of summands, we apply the CLT for compound Poisson processes. As $\lambda \rightarrow \infty$:

$$\frac{L_i - \mathbb{E}[L_i]}{\sqrt{\text{Var}(L_i)}} \xrightarrow{d} \mathcal{N}(0, 1) \tag{12}$$

Step 4: Self-averaging of aggregate losses. Consider the system-wide average loss $\bar{L} = \frac{1}{N} \sum_{i=1}^N L_i$. Since the L_i are identically distributed with mean $\mu_L = \lambda q\mu$ and variance $\sigma_L^2 = \lambda q(\sigma^2 + \mu^2)$, the Law of Large Numbers gives:

$$\bar{L} \xrightarrow{a.s.} \mu_L = \lambda q\mu \quad \text{as } N \rightarrow \infty \tag{13}$$

This is the self-averaging property: in the large- N limit, the realized aggregate loss concentrates deterministically around its expectation, and fluctuations vanish. The probability of an extreme aggregate loss event decays as:

$$\mathbb{P}(\bar{L} > \mu_L + t) \sim \exp\left(-\frac{t^2}{2\sigma_L^2/N}\right) \quad \text{as } N \rightarrow \infty \quad (14)$$

confirming that tail events are exponentially suppressed. The Gai-Kapadia model is therefore structurally incapable of generating the fat-tailed aggregate loss distributions observed in real financial crises.

A.3 When the CLT Fails: The Power Law Regime

We now show that replacing the finite-variance assumption on w_{ij} with a Power-Law distribution invalidates both the CLT and the self-averaging result.

Suppose w_{ij} follows a Pareto distribution with tail exponent α :

$$f_W(w) = \frac{\alpha x_{\min}^\alpha}{w^{\alpha+1}}, \quad w \geq x_{\min}, \quad \alpha > 0 \quad (15)$$

For $1 < \alpha \leq 2$, the mean $\mathbb{E}[w_{ij}]$ is finite but the variance $\text{Var}(w_{ij})$ is infinite. For $\alpha \leq 1$, even the mean diverges. In either case, the conditions required for equation (8) to hold are violated.

The standard CLT requires finite variance. When $\text{Var}(w_{ij}) = \infty$, the Generalized Central Limit Theorem applies instead [13]: the normalized sum of i.i.d. Pareto variables converges not to a Gaussian but to a **Lévy α -stable distribution** S_α , characterized by power-law tails of the same exponent α .

The practical consequence is stark. In the Gaussian regime, self-averaging holds and aggregate losses concentrate around the mean:

$$\bar{L} \xrightarrow{a.s.} \mu_L \quad (\text{Gaussian world}) \quad (16)$$

In the Power-Law regime, the aggregate loss is instead dominated by its largest term:

$$\bar{L} \approx \frac{1}{N} \max_i L_i \quad (\text{Power-Law world}) \quad (17)$$

This result — that the maximum single loss dominates the total — is a direct consequence of the heavy tail. It means that the failure of one sufficiently large institution can account for a macroscopic share of the system's total stress, regardless of how many other institutions are present. Self-averaging fails completely, and the Law of Large Numbers provides no protection.

This mathematical shift furnishes the rigorous explanation for the empirical observation that motivated this paper: a single institution — Lehman Brothers — could destabilize a global financial system that appeared statistically stable under Gaussian assumptions. The system was not stable; it was merely operating in a regime where the dominant risk was invisible to the tools being used to measure it.